

Abd-Ur-Rehman

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EDUCATION

University of Toronto

Toronto, ON

HBSc, Computer Science Specialist (Co-op) – Software Engineering

Expected Graduation: Aug. 2027

- Coursework: Software Engineering, Data Structures & Algorithms, Software Design, Systems Programming, OOP.
- Activities: Hack the Valley X Hackathon Participant (Oct. 2025)

EXPERIENCE

Software Engineer – Cloud & AI/ML (Co-op)

Jan. 2026 – Apr. 2026

Nokia

Ottawa, ON

- Reduced query resolution time from 2–3 min to 30–40s on the **aApp Support Agent** by co-architecting a multi-agent AI pipeline (Supervisor, Planning, 4 domain agents) in a 2-person team, integrating ChromaDB, Kubernetes, Redis, and Gitea via MCP.
- Cut repeated LLM inference costs by ~60% by implementing a two-tier Redis semantic cache (global + thread-scoped) with cosine-similarity lookup, achieving 1–2s responses on cache hits across sessions.
- Streamlined developer onboarding by exposing the multi-agent workflow as a REST API with LangGraph checkpointing for persistent conversation threading, reducing integration time by ~50%.
- Scaled the architecture into a generalized **Telecom Support Agent** for Nokia's 5G core network functions by replicating and adapting the multi-agent pipeline, delivering a production-ready second agent within 2 weeks.

Software Engineer – AI Automation & Integration (Co-op)

Sep. 2025 – Dec. 2025

Digital Technology Canada (DTCForce)

Oakville, ON

- Automated financial data retrieval across 41 workflows by building the **AI Financial Agent** in a 3-person team, orchestrating OAuth-secured APIs with n8n + LLMs.
- Reduced workflow turnaround from 5–10 min to 1–2 min by developing AI-driven n8n automations in JavaScript, integrating third-party APIs and backend services for multi-step orchestration.
- Cut average response latency by 80% by iterating on routing logic, formatting pipelines, and error-handling approaches across end-to-end workflows.
- Secured stakeholder buy-in by presenting live demos of AI workflows, translating technical implementations into clear business value and driving adoption of 3 new automation pipelines.

PROJECTS

Redistribution Map Platform | *FastAPI, React, PostgreSQL* | [GitHub](#)

May 2026 – Aug. 2026

- Surfaced which of Canada's 338 electoral ridings generated the most public commentary by building an interactive MapLibre/PMTiles heatmap with click-to-popup breakdowns and Keycloak OIDC authentication.
- Increased test coverage from 66% to 100% (backend) and 46% to ~99.9% (frontend) across 500+ tests by writing test suites for a team codebase, uncovering and fixing real production bugs including a silently no-op async delete method.
- Eliminated data-integrity bugs across 3 submission types by tracing and fixing foreign-key mismatches between internal and public district identifiers directly against the live schema.

PlanetZe | *Java, Android Studio, Firebase* | [GitHub](#) | [Demo](#)

Nov. 2024 – Dec. 2024

- Delivered a carbon-footprint tracking Android app in a 6-person Agile team by developing core app structure, navigation, and reusable UI components in Java and XML using Android Studio.
- Powered 100% of personalized tracking features by building and integrating an XML-based questionnaire that gathered sustainability data, with responsive UI layouts using Canva assets to improve user engagement.
- Achieved sub-second data latency by implementing Firebase Realtime Database integration to store and sync survey responses, ensuring reliable cross-device data access.

Linux FD Table Monitor | *C, Linux, Bash, Git* | [GitHub](#)

Feb. 2025 – Mar. 2025

- Enabled real-time visibility into file descriptor usage across all user processes by building a C-based Linux monitoring tool that traverses `/proc` with `readdir()`, `readlink()`, and `stat()` syscalls.
- Supported 5+ CLI flags and modular functions (per-process, system-wide, vnode, composite, summary, threshold, export) by designing a flag-based argument parser with configurable output targets.
- Enhanced data export performance by 30% by implementing custom binary serialization alongside text export, benchmarked with shell scripts across single-PID and system-wide scans.

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, TypeScript, SQL, C, HTML, CSS

AI/ML: LangGraph, LangChain, ChromaDB, MCP (Model Context Protocol), LLMs, ReAct Agents, RAG, Semantic Caching, Prompt Engineering, Pydantic

Frameworks: React, Next.js, Node.js, Express.js, FastAPI, Tailwind CSS, Firebase

Cloud & DevOps: Docker, Helm, OpenShift, Kubernetes, Redis, MongoDB, AWS, GCP, CI/CD, Git/GitHub, Linux

Practices & Tools: Agile/Scrum, REST APIs, OAuth 2.0, Jira, VS Code, n8n Automation